PROGRAM 4 – CRYPT-ARITHMETIC PROBLEM

AIM

To write a Python program to solve the Crypt-Arithmetic problem using permutations.

ALGORITHM

1. Consider the equation SEND + MORE = MONEY.
2. Assign a unique digit to every alphabet.
3. Generate possible digit combinations using permutations.
4. Ensure that the first letters S and M are not zero.
5. Convert the words into numerical values.
6. Check whether SEND + MORE = MONEY.
7. Display the valid digit assignment.

CODE

```
from itertools import permutations
letters="SENDMORY"
found=False
for values in permutations(range(10),8):
    s,e,n,d,m,o,r,y=values
    if s==0 or m==0:
        continue
    SEND=1000*s+100*e+10*n+d
    MORE=1000*m+100*o+10*r+e
    MONEY=10000*m+1000*o+100*n+10*e+y
    if SEND+MORE==MONEY:
        print("SEND =",SEND)
        print("MORE =",MORE)
        print("MONEY =",MONEY)
        found=True
        break
if not found:
    print("No solution found")
```

INPUT:

No input required.

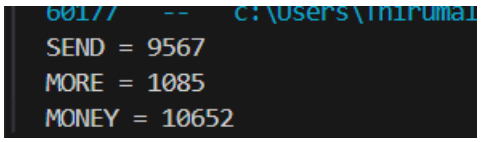
The program automatically solves:

SEND

+ MORE

MONEY

OUTPUT:



RESULT

Thus, the Crypt-Arithmetic problem was successfully solved using permutations in Python.